



Impacts of forest management intensity on carbon accumulation of China's forest plantations



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ABSTRACT

Characterized by young age and a large area been newly planted, China's forests were projected to have high carbon (C) uptake potential in the future. While such a potential assumes continuous human management, **the effects of forest management on C uptake potential are less clear**. Using surveyed data collected from a recent nationwide field campaign conducted between 2011 and 2015, we examined human management effects on C stock dynamics in three of the most widely planted forests in southern China, i.e. *Pinus massoniana*, *Cunninghamia lanceolata*, and *Eucalyptus*. Vegetation C accumulation rates were larger under higher management intensities in *Pinus massoniana* and *Eucalyptus*, while there was no such a pattern in *Cunninghamia lanceolata*. Human management has contributed to vegetation C increment by 2.42 Tg C yr⁻¹ and 0.55 Tg C yr⁻¹ in *Pinus* and *Cunninghamia* forests. While in comparison, human management impacts on C accumulation were not accountable in *Eucalyptus* due to lack of naturally regenerated forests. Moreover, we found that potential vegetation C accumulation can be enhanced by 7.86 Tg C yr⁻¹ from higher management intensity with 3.86, 2.17, and 1.83 Tg C yr⁻¹ from *Pinus massoniana*, *Cunninghamia lanceolata*, and *Eucalyptus*, respectively. We also advise that different management intensities should be considered to avoid biases for projection of future C sink from forest plantations regionally or nationally.

1. Introduction

Forest carbon (C) stock accounts for about 45–54% of the total C in terrestrial ecosystems (Bonan, 2008; Carvalhais et al., 2014), and the annual C uptake from global forests could offset about 26% of the C emission from fossil fuel combustion (Pan et al., 2011). Previous studies quantified forest C stock and its dynamics in China (Fang et al., 2001, 2014, Piao, 2005, He et al., 2017, Zhao et al., 2019), given that China's forest plantations are about a quarter of the total global forest plantations (FAO, 2010; Peng et al., 2014). Traditional estimation of forest C stock and its change involves biomass harvesting and soil sampling, which was however, impossible to be performed at regional and national scales. Thus, researchers have developed allometric equations for large scale estimation of C stock by establishing a relationship between the various physical parameters obtained from sampled field plots and forest inventory data (Fang et al., 2001, Piao, 2005, He et al., 2017, Zhao et al., 2019). For example, Fang et al. (2001) estimated that China's planted forest C stock increased from 0.27 Pg in 1970s to 0.72 Pg in 1998 using national forest inventory data. Pan et al. (2004)

further developed age-group based equations for forest C estimations, and found that neglecting age-group effects may cause 27% and 89% of the differences in C pools and C fluxes estimations.

Although these studies greatly expanded our knowledge in understanding C stock and its dynamic in China's forests, less attention has been paid to the human management effects on forest C accumulation. Nonetheless, this should be investigated for potential enhancement of C uptake from optimized forest management, which is particularly important for China's subtropical forests identified as an essential C sink (Yu et al., 2014). Regional and national estimations of forest management impacts, however, were generally rarely found due to a lack of information for such a large scale. Recently, a nationwide field campaign was conducted between 2011 and 2015 across China, following consistent field inventory protocol and laboratory methodology (Tang et al., 2018). Specifically, human managements were collected and recorded during the field survey, including management type and management intensity (Tang et al., 2018). This allowed the quantification of human management effects on C accumulation in plantations at large scale. In this study, we focus on the three major planted forests

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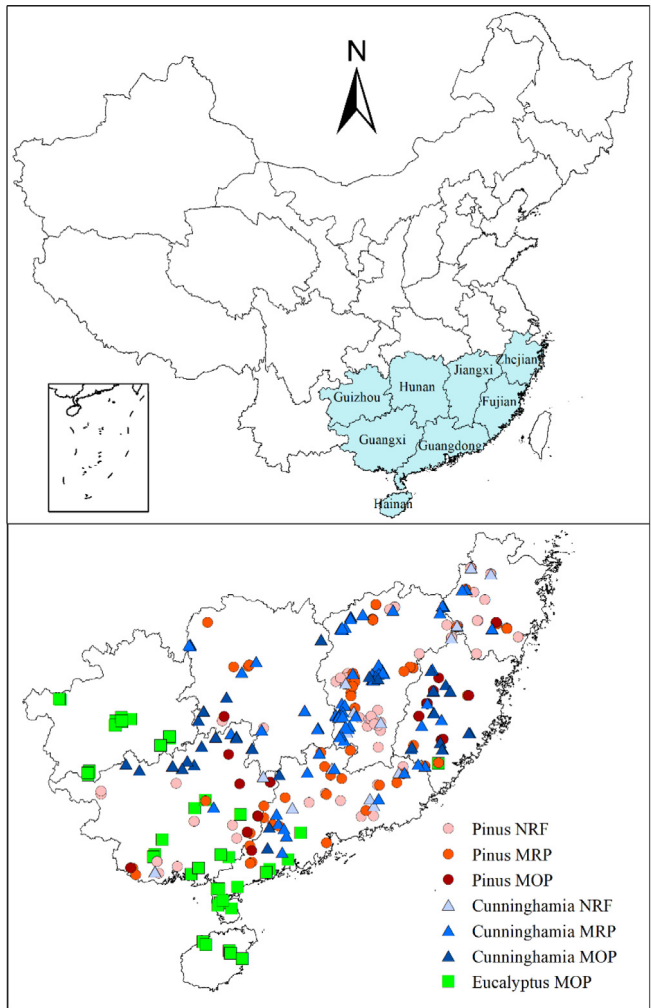


Fig. 1. Distribution of the sites collected from the field campaign in the study area.

in southern China (i.e. *Pinus massoniana*, *Cunninghamia lanceolata*, and *Eucalyptus*) and aim to 1) quantify C dynamics in plantations under different management intensities; and 2) project potential biases in C dynamic estimations when not including human management effects.

2. Data and methods

2.1. Study area

During a nationwide field campaign conducted between 2011 and 2015, a total of 35,800 sites were randomly sampled with 2748 from planted forests. In this study, we examined all available sites collected from *Pinus massoniana* (*Pinus* hereafter), *Cunninghamia lanceolata* (*Cunninghamia* hereafter), and *Eucalyptus* (Fig. 1). The vegC and SOC for each sites have been estimated by Tang et al. (2018). Sites with missing

age, vegetation C (vegC), or soil C were excluded from the analysis, and 530 sites in total were retained for further analysis. The field sampling recorded information of geographical locations, soil properties, vegetation properties, and human managements. Specifically, human managements, including management type and management intensity, were recorded by interviewing local residents or officers of agriculture and forestry departments (Tang et al., 2018). The forest management types include timber harvesting, thinning, understory plants harvesting, plantation, fertilizing, and grazing. However, details of management implemented in each plantation (e.g. frequency, timing) were not available. Thus, in this study, sites without human activities or with slight impacts were flagged lower-managed ones; and otherwise they were flagged as higher-managed ones.

In this study, we focused on forest sites under three different management groups, i.e. naturally regenerated forests (NRFs), mixed regenerated plantations (MRPs), and monoculture plantations (MOPs). MOPs are plantations intensively managed to maintain a single tree species. In comparison, MRPs are forests planted and less intensively managed, which therefore have other tree species naturally regenerated into either canopy or shrub layer. The NRFs were barely/slightly disturbed by human intervention, and was tagged as “natural succession” during field survey. Therefore, the human management intensity increases from NRFs to MRPs and to MOPs. In this study, each of the three management groups were further divided into two management levels based on the human management intensity recorded in each plot (see Table 1). Thus, in total six management intensity levels were identified with the management intensity rank of $NRF1 < NRF2 < MRP1 < MRP2 < MOP1 < MOP2$ (Table 1). The total number of sites sampled was 222 from *Pinus* (102 for NRFs, 79 for MRPs, and 41 for MOPs), 225 from *Cunninghamia* (20 for NRFs, 147 for MRPs, and 58 for MOPs), and 83 for *Eucalyptus* (16 from MOP1, and 67 from MOP2).

2.2. Data processing

Pinus, *Cunninghamia*, and *Eucalyptus* plantations account for 14.18%, 40.54%, and 13.48% of the total planted forests in the study area, respectively (Division of forest resources management, 2010). According to the classification of forest age for different forest types (Xiao, 2005), the sampled forest plots cover young-aged, middle-aged, pre-mature, mature, and over mature groups for *Pinus* and *Cunninghamia*, while *Eucalyptus* plantations were all in young-aged and middle-aged groups. Linear regression was applied for each of the management groups in each forest types to examine C stock change along forest age gradients. The level of statistical significance was set a priori at $\alpha = 0.05$. Data were analyzed with Sigmaplot (v14.0).

3. Results

3.1. Carbon stock change with forest age in *Pinus massoniana* forests

VegC was found significantly increase with age in *Pinus* forests under all the six levels of management intensities in the three management groups, i.e. NRFs, MRPs and MOPs (Fig. 2). Among the three management groups, vegC increase with age was largest in MOPs ($357.32 \text{ g C yr}^{-1}$, $p < 0.05$, Fig. 2c), followed by MRPs (240.05 g C

Table 1
Six management intensity levels for forest plantations.

Management groups	Vegetation composition	Management intensity levels	
		lower	higher
Naturally regenerated forests (NRFs)	Other woody species* regenerated from sites tagged as natural succession	NRF1	NRF2
Mixed regenerated plantation (MRPs)	Other woody species regenerated from managed plantations	MRP1	MRP2
Monoculture plantation (MOPs)	No other woody species in managed plantations	MOP1	MOP2

* Other woody species are woody species that were not planted initially.

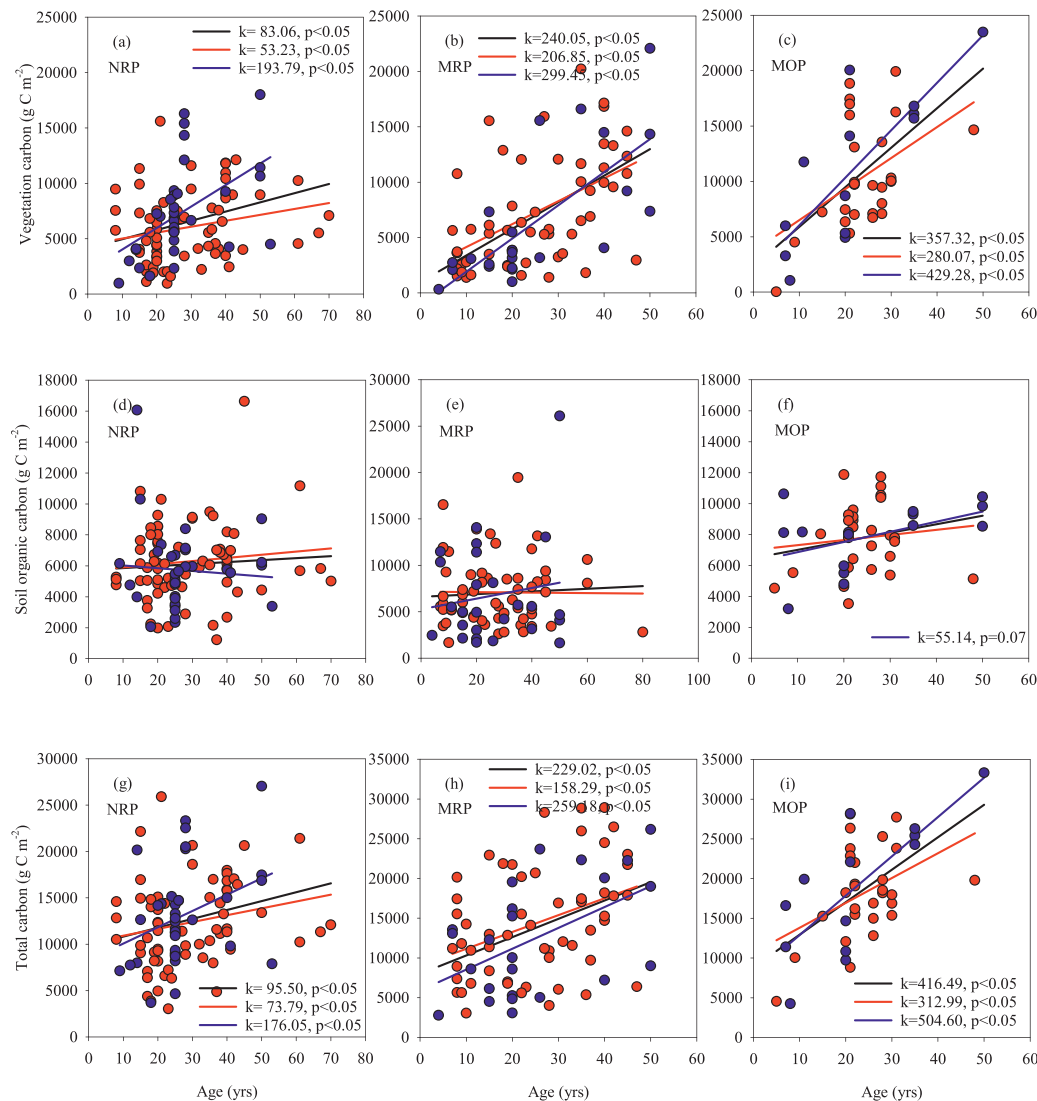


Fig. 2. Forest vegetation carbon, soil carbon and total carbon stock changes along age gradients in naturally regenerated forests (NRFs; a, d, g), mixed regenerated plantations (MRPs; b, e, h), and monoculture plantations of *Pinus massoniana* forests (MOPs; c, f, i). (k: slope of the linear regression line; upper panel: vegetation carbon; mid-panel: soil carbon; lower panel: total carbon; circles with red and blue colors indicate lower- and higher-intensity management levels in each group, respectively; red, blue, and black lines indicate the regressions for lower-management, higher- management, and pooled both lower- and higher-management samples in each level of management group). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

yr^{-1} , $p < 0.05$, Fig. 2b) and NRFs ($83.06 \text{ g C yr}^{-1}$, $p < 0.05$, Fig. 2a). In comparison, soil organic carbon (SOC) was only slightly and positively correlated with forest age in higher management level of MOPs (MOP2, $55.14 \text{ g C m}^{-2} \text{ yr}^{-1}$, $p = 0.07$, Fig. 2f), while no significant relationship was detected in NRFs and MRPs (Fig. 2d, e). Similar to vegC, total C increased with forest age in MOPs ($416.49 \text{ g C m}^{-2} \text{ yr}^{-1}$, $p < 0.05$, Fig. 2i), MRPs ($229.02 \text{ g C m}^{-2} \text{ yr}^{-1}$, $p < 0.05$, Fig. 2h), and NRFs ($95.50 \text{ g C m}^{-2} \text{ yr}^{-1}$, $p < 0.05$, Fig. 2g).

In each of the three management groups, a larger C increment rate was identified in the higher management intensity level (Fig. 2). For vegC, the higher management level in NRFs (i.e. NRF2, the blue line in Fig. 2a, $193.79 \text{ g C yr}^{-1}$, $p < 0.05$) showed a larger increase rate comparing with lower management level NRFs (NRF1, the red line in Fig. 2a, $53.23 \text{ g C yr}^{-1}$, $p < 0.05$). Similar patterns were found in vegC of MRPs (299.45 vs $206.85 \text{ g C yr}^{-1}$) and MOPs (429.28 vs $280.07 \text{ g C yr}^{-1}$), and total C of NRFs (176.05 vs $73.79 \text{ g C m}^{-2} \text{ yr}^{-1}$), MRPs (259.18 vs $158.29 \text{ g C m}^{-2} \text{ yr}^{-1}$) and MOPs (504.60 vs $312.99 \text{ g C m}^{-2} \text{ yr}^{-1}$).

3.2. Carbon stock change with forest age in *Cunninghamia lanceolata* forests

In *Cunninghamia* plantations, only five management intensity levels were available for comparing (Fig. 3). We found that vegC was also significantly increased with increasing age in each of the management levels of *Cunninghamia* plantations (Fig. 3a, b, c). For MRPs and MOPs, increment rates were relatively small between each of the two management levels (Fig. 3b, c). Nonetheless, across the three groups, we found a similar pattern as *Pinus* forests that vegC increment rate was largest in MOPs ($230.67 \text{ g C m}^{-2} \text{ yr}^{-1}$), followed by NRFs ($198.72 \text{ g C m}^{-2} \text{ yr}^{-1}$, $p < 0.05$) and MRPs ($180.50 \text{ g C m}^{-2} \text{ yr}^{-1}$, $p < 0.05$). Conversely, SOC in *Cunninghamia* forests was only slightly and positively correlated with forest age in MOPs, but this correlation was only slightly significant ($47.34 \text{ g C m}^{-2} \text{ yr}^{-1}$, $p = 0.07$, Fig. 3f). Likewise, no significant relationship was detected in forest age and SOC in NRFs and MRPs (Fig. 2d, e). Surprisingly, total C change along forest age gradient was not significant in NRFs ($p = 0.18$, Fig. 3g), while it was significant in MRPs ($229.02 \text{ g C m}^{-2} \text{ yr}^{-1}$, $p < 0.05$, Fig. 3h) and

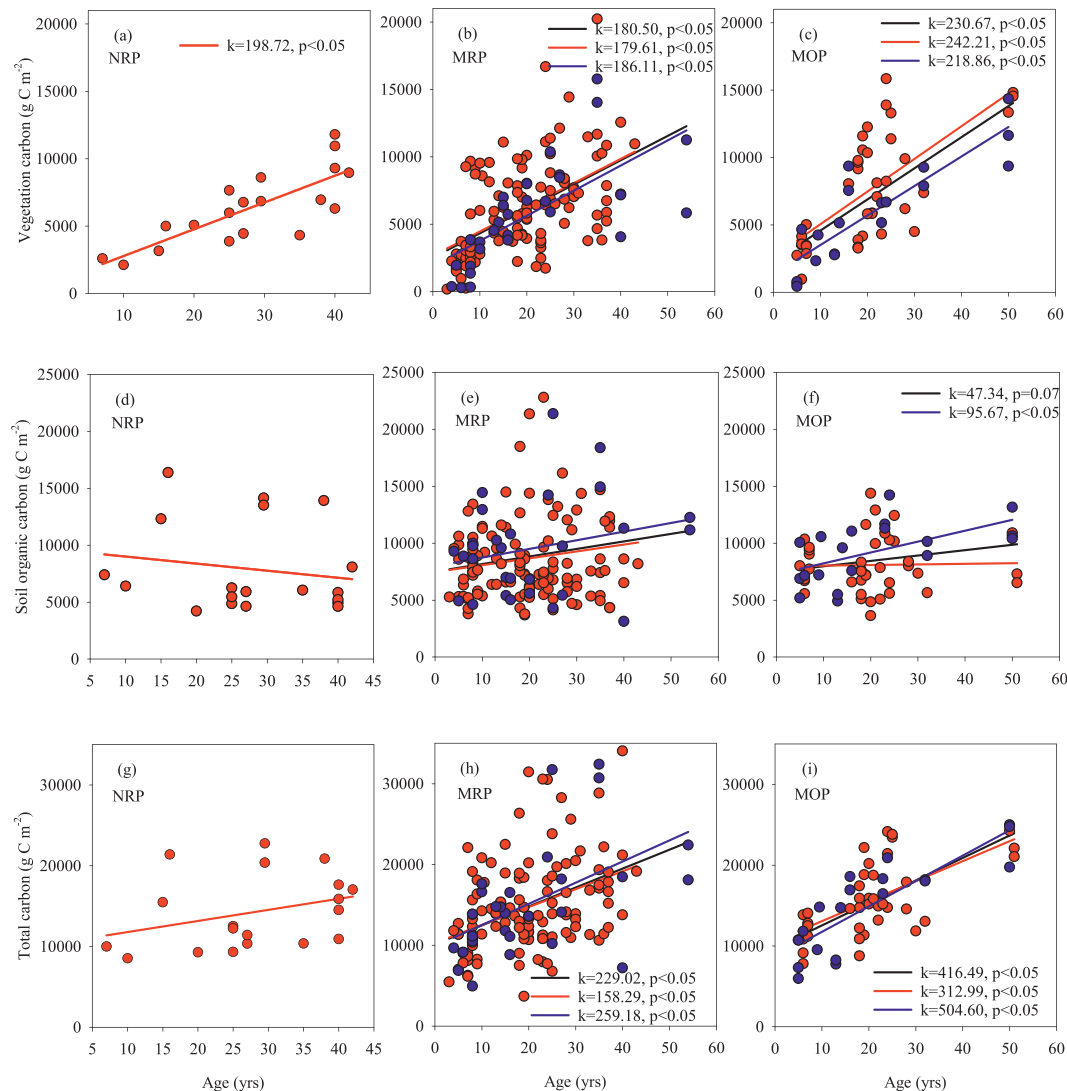


Fig. 3. Forest vegetation carbon, soil carbon and total carbon stock changes along age gradients in naturally regenerated forests (NRFs; a, d, g), mixed regenerated plantations (MRPs; b, e, h), and monoculture plantations (MOPs; c, f, i) of *Cunninghamia lanceolata* forests. (Upper panel: vegetation carbon; mid-panel: soil carbon; lower panel: total carbon; circles with red and blue colors indicate lower- and higher-intensity management levels in each group, respectively; red, blue, and black lines indicate the regressions for lower-management, higher- management, and pooled both lower- and higher-management samples in each level of management group). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

MOPs ($416.49 \text{ g C m}^{-2} \text{ yr}^{-1}$, $p < 0.05$, Fig. 3i).

For MRPs, a slightly larger C increment with age was identified in the higher management intensity level (MRP2, blue line in Fig. 3b), while no significant relationship was found between SOC and age, leading to a higher total C in MRP2 than MRP1 (Fig. 3h). In comparison, in MOPs, a larger C increment with age was identified in the lower management intensity type (MOP1, red line in Fig. 3c, $242.21 \text{ g C m}^{-2} \text{ yr}^{-1}$). However, SOC increased with age in the higher intensity management of MOPs (MOP2, Fig. 3f, $95.67 \text{ g C m}^{-2} \text{ yr}^{-1}$, $p < 0.05$), but no significant relationship was detected in lower intensity management of MOPs (MOP1), rendering to a higher total C increment rate with age in MOP2 (Fig. 3i, 504.60 vs $312.99 \text{ g C m}^{-2} \text{ yr}^{-1}$).

3.3. Carbon stock change with forest age in *Eucalyptus* forests

Surveyed *Eucalyptus* plantations were all monocultures (Fig. 4). The VegC in *Eucalyptus* plantations increased significantly with increasing age under MOP2 ($919.03 \text{ g C m}^{-2} \text{ yr}^{-1}$, $p < 0.05$, Fig. 4a), a rate that is over two-fold the increasing rate found in MOP1

($477.09 \text{ g C m}^{-2} \text{ yr}^{-1}$, $p = 0.09$, Fig. 4a). In comparison, total C was found to significantly increase with age in MOPs ($609.42 \text{ g C m}^{-2} \text{ yr}^{-1}$, $p = 0.05$, Fig. 4c).

3.4. Stem densities in different forests

Highest stem densities were found in *Eucalyptus* forests at around 2290.11 – 3487.80 stems ha^{-1} , followed by *Pinus Massoniana* (612.19 – 1662.52 stems ha^{-1}) and *Cunninghamia lanceolata* forests (871.69 – 1941.32 stems ha^{-1} , Table 2). The largest variation of stem densities between sites was also found in *Eucalyptus* forests, while the variations were similar in *Pinus Massoniana* and *Cunninghamia lanceolata* forests (Table 2). Generally, the stem densities were higher under higher management levels in each management groups, except MRPs in *Pinus Massoniana* and *Cunninghamia lanceolata* forests (Table 2).

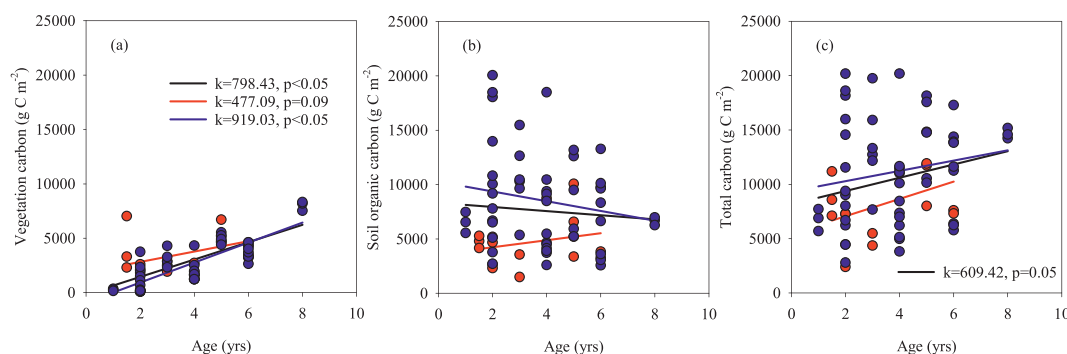


Fig. 4. Change of (a) vegetation carbon, (b) soil organic carbon, and (c) total carbon with increasing age in *Eucalyptus* plantations (circles with red and blue colors represent lower- and higher-intensity management levels, respectively; red, blue, and black lines indicate the regressions for lower-management, higher- management, and pooled both lower- and higher-management samples). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 2

Stem densities in each forest types under different management intensity levels.

Forest types	Management intensities	Stem densities (stems ha ⁻¹)	Standard deviations
<i>Pinus massoniana</i>	NRF	NRF1	1380.85
		NRF2	1662.52
	MRP	MRP1	1555.98
		MRP2	1509.89
	MOP	MOP1	612.19
		MOP2	1026.89
<i>Cunninghamia lanceolata</i>	NRF	–	1941.32
	MRP	MRP1	1910.76
		MRP2	1848.79
	MOP	MOP1	871.69
		MOP2	1743.95
			930.54
<i>Eucalyptus</i>	MOP	MOP1	2290.11
		MOP2	3487.80
			3652.59
			5344.01

4. Discussion

4.1. Carbon stock change with forest age

vegC was found consistently and significantly increased with increasing age in all management levels in three tree species. For *Pinus* forests, the average vegC accumulation rate in NRFs was 83.06 g C m⁻² yr⁻¹. This rate is close to the rates of 95.33 and 107.57 g C m⁻² yr⁻¹ (unknown management intensity) grew from 10 yrs to 40 yrs, reported by Xu et al. (2010), and the natural *Pinus* forests grew from 18 yrs to 63 yrs, reported by Pan et al. (2014). However, an exception was also found by Guo et al. (2018), who reported an woody vegC increment at 244 g C m⁻² yr⁻¹ from 10.35 Mg ha⁻¹ (16 yrs) to 83.55 Mg ha⁻¹ (46 yrs) from a NRFs in Jiangxi Province. Such a high increment rate is close to the vegC increment of MRPs derived in this study. For MRPs, the forest vegC increment rate of 240.05 g C m⁻² yr⁻¹ derived from this study is close to the range of 245–266 g C m⁻² yr⁻¹ found by Wei and Ma (2007) and Han et al. (2017), assuming 25 yrs of growth from mid-age forest to mature forest in *Pinus* MRPs from southern China. In comparison, the average *Pinus* vegC accumulation rate from MOPs (357.32 m⁻² yr⁻¹) is in the range of 276–778 g C m⁻² yr⁻¹ reported in previous studies with intensively managed *Pinus* forest farms (Fang et al., 2003, Qin et al., 2011, Tao et al., 2011, 2012).

Surprisingly, we found that vegC increment rate in *Pinus* forest was over eight-times higher in plantations under highest management level (NRF1, 429.28 g C m⁻² yr⁻¹; Fig. 2) than plantations under the lowest management level (MOP2, 53.23 g C m⁻² yr⁻¹; Fig. 2). Similarly, the

total C changes showed the same pattern despite the different increasing rates found between management intensity levels (Fig. 2), indicating that total C change in *Pinus* forest was mainly determined by vegC increase. Such great differences of C increment rates between management levels indicate that human management intensity plays a pivotal role in C accumulation in *Pinus* forests. By comparing the three management groups in *Pinus* forests, switching from naturally regenerated forests (NRFs, barely managed) to mixed regenerated plantations (MRPs, managed) greatly increases vegC accumulation from 83.06 to 240.05 g C m⁻² yr⁻¹, implying large benefits from forest tending and protection. Moreover, since the sampling process followed a random design, the management intensities from sampled plantations in each forest type are considered representative for the region. Simply extrapolating the results obtained to convert all NRFs to MRPs in *Pinus* forests will increase annual vegC accumulation by 38.05% in the study area. Although not realistic and more aggressive, conversion of all NRFs and MRPs to MOPs could increase annual vegC accumulation by 88.48% in the study area. However, MOPs has been criticized for low ecosystem resiliency and high tree mortality due to environmental stresses across the planted regions (Zhang et al., 2000, Wang et al., 2007, Xu, 2011, Liang et al., 2016a, Yu et al., 2019). Besides, despite its importance, C sequestration is not the only benefit targeted in China's forest plantations. For example, Grain to Green Program was aiming at soil erosion reduction benefit, and Natural Forest Conservation Program also highlighted water retention and flood control benefits (Liu et al., 2008). Therefore, a more attainable potential in C stock enhancement is to adapt tending and protection strategies in under-managed forests suitable for local needs.

For *Cunninghamia* forests, vegC differences of within- and between-manage groups were much smaller than for *Pinus* forest. Conversely to *Pinus* forest where total C change was dominated by vegC change, SOC in *Cunninghamia* forests played an important role in determining total C change (Fig. 3). For example, although the vegC was significantly increased with age in NRFs, MRPs, and MOPs groups, the large variation in SOC resulted into non-significant change of total C along the age gradients (Fig. 3). Moreover, annual vegC increments were relatively small between the two management levels in each of the MRPs (179.61 vs 186.11 g C m⁻² yr⁻¹; Fig. 3b) and MOPs (242.21 vs 218.86 g C m⁻² yr⁻¹; Fig. 3c) groups. Liang et al. (2016b) showed that vegC growth from young-age to middle-age was at 210.05 g C m⁻² yr⁻¹ in *Cunninghamia* forests. Similarly, Lan et al. (2016) surveyed 45 plots in southern China (Guangxi Province), and reported vegC (wood only) increased from 33.51 (6 yrs) to 133.75 (50 yrs) Mg C m⁻² yr⁻¹, equivalent to a growth rate at 227.82 g C m⁻² yr⁻¹. Thus, our result of the average vegC incensement rates of 180.50–230.67 g C m⁻² yr⁻¹ was close to the reported values.

Surprisingly, due to the contribution from SOC change, total C increment rates were differed between the two management levels in

both MRPs (158.29 vs 259.18 g C m⁻² yr⁻¹; Fig. 3b) and MOPs (312.99 vs 504.60 g C m⁻² yr⁻¹; Fig. 3c) groups. Shi et al. (2017) examined forest vegC growth and SOC change in *Cunninghamia* forests under natural regeneration (NRFs) and human management (MRPs) approaches. Their study revealed that NRFs was higher in vegC accumulation, while MRPs better promoted SOC increment. This is consistent to our results that vegC increase rate was lower in NRFs (198.72 g C m⁻² yr⁻¹, Fig. 3) than in MRPs (180.50 g C m⁻² yr⁻¹, Fig. 3), although the SOC changes were not significant in both management groups. However, we found that SOC increased significantly in MOPs (47.34 g C m⁻² yr⁻¹), while it was majorly contributed from highest management levels of MOP2 (95.67 g C m⁻² yr⁻¹). This is consistent with the results of Liu et al. (2017), who found that forest tending increased SOC by 330 g C m⁻² in young *Cunninghamia* forests. Thus, if intensively and properly managed, a large potential of C uptake in both vegetation biomass and soil can be expected in *Cunninghamia* plantations.

Eucalyptus is one of the most fast-growing and intensively managed tree types in China (Xie, 1998). The reported vegC increase rates largely varied across studies and forest ages. For example, Wen et al. (2014)'s study showed that annual C fixed in vegetation biomass ranged from 310 to 1258 g C m⁻² yr⁻¹ for *Eucalyptus* aged from 1 to 8 yrs. Tao et al. (2011) even found that *Eucalyptus* vegC increased from 2.68 to 31.62 Mg C m⁻² yr⁻¹ in two years, which is equivalent to 1447 g C m⁻² yr⁻¹ in vegC increase. In this study, we found that annual vegC growth rate ranged from 477.09 to 919.03 g C m⁻² yr⁻¹ under two levels of management intensities with an average of 798.43 g C m⁻² yr⁻¹ (Fig. 4), which is about two- to three-folds of the C increase rate found in *Pinus* and *Cunninghamia* plantations. In comparison, SOC in *Eucalyptus* plots largely varied with ages. A similar result was also reported by Zhu et al. (2018), who showed that SOC largely varied (8036–11321 Mg C m⁻²) across ages (8, 10, and 15 yrs), although the result was statistically non-significant. This may be because the trees were very young and SOC were majorly determined by the initial status in each plantation, while SOC accumulation contributed from planted trees was slow and may not be observed in such a short time. Besides, SOC was also reported to decrease for the first few years after planting (Grigal and Berguson, 1998, Turner and Lambert, 2000, Paul et al., 2003), which is consistent with the pattern observed in this study (Fig. 4b).

4.2. Implications for forest management

The three planted forests analyzed in this study were the major species planted in the study area. According to the Division of Forest Resources Management (DFRM, 2010) report, *Pinus*, *Cunninghamia*, and *Eucalyptus* plantations account for 14.18% (2.29 Mha), 40.54% (6.65 Mha), and 13.48% (2.18 Mha) of the planted area in the provinces covered in this study. Thus, the area ratio of *Pinus*, *Cunninghamia*, and *Eucalyptus* plantations were 21:59:20 derived from the DFRM report, which is close to 29:50:20 from the sampled plantations, indicating the field campaign is representative for the region. Assuming the three management groups from sampled plantations were representative for the region, 46%, 36%, and 18% of *Pinus* plantations were managed in NRFs, MRPs, and MOPs groups in southern China. While in comparison, 9%, 65%, and 26% of *Cunninghamia* plantations were managed in NRFs, MRPs, and MOPs groups, respectively. Therefore, human management have contributed to vegC by 2.42 Tg C yr⁻¹ and 0.55 Tg C yr⁻¹ in *Pinus* and *Cunninghamia* forests.

Estimated potential vegC accumulation can be enhanced by 2.89 Tg C yr⁻¹ and 0.97 Tg C yr⁻¹ if *Pinus* plantations are converted from NRFs and MRPs to MOPs in the study area since the 7th national forest inventory. Despite that *Cunninghamia* plantations did not show a consistent increase of vegC from NRFs to MRPs and MOPs (Fig. 3), potential vegC accumulation can still be enhanced by 2.17 Tg C yr⁻¹ if *Cunninghamia* plantations are converted from MRPs and MOPs. In

comparison, *Eucalyptus* plantations were all managed as MOPs, but 19% of the plantations were managed in MOP1 (lower management intensity level). Thus, an additional 1.83 Tg C yr⁻¹ can be obtained by increasing management levels from MOP1 to MOP2 in *Eucalyptus* plantations. Therefore, in total of 7.86 Tg C yr⁻¹ was available from higher management levels of the three plantations types in the study area. This increment helps to enhance C uptake at about 11% for East Asian Monsoon subtropical forests (Yu et al., 2014). When comparing with national C uptakes from China's forests, such increment amounts to 4.6–9% of the annual forest C uptake capacity derived from Xu et al. (2010) and Yao et al. (2018).

In comparison, SOC change was much more complex and heterogeneous than vegC change as the former closely related to land use history (Paul et al., 2002, Don et al., 2011, Poeplau et al., 2011). Aside from the *Eucalyptus* plantations that were generally very young, we also found that SOC were only positively and significantly related to age in MOP2 of *Pinus* and *Cunninghamia* forests. For NRFs and MRPs, SOC changes were not significant as forests grew. Thus, for plantations targeting for C accumulation, intensive management may be required for enhancing both above- and below-ground C in *Pinus* and *Cunninghamia* forests.

Results from this study revealed that different management intensities should be considered for projection of future C sink from forest plantations regionally or nationally. For example, using biased sampled plots, such as plots majorly sampled from intensively managed forest farm or timber forests, will greatly overestimate forest C accumulation rate. On the contrary, if the samples were mainly collected from naturally regenerated plots, forest C sink maybe largely underestimated when extrapolating plot-level C accumulation rates to regional or national estimates. For example, the forest C accumulation rates of *Pinus*, *Cunninghamia*, and *Eucalyptus* in the study of Xu et al. (2010) were close to the results obtained from NRFs in this study. Therefore, using pooled samples collected from NRFs, MRPs, and MOPs from this study will provide a much higher estimate of C uptake potential.

Note that stem density differed between management intensity levels, which also affected C accumulation rates. For both *Pinus Massoniana* and *Cunninghamia lanceolata* forests, stem densities were much lower in MOP than in MRP and NRF, implying human management (e.g. thinning) may help reduce the density while enhancing C accumulations. This is understandable as stem density is largely determined by initial planting density and subsequently thinning practices.

4.3. Uncertainties

There are uncertainties that may affect the results of this study. For instance, the planting density (stem density), climate, topography differences of the each species in each plot will influence the C accumulation rate as tree grows. Long-term monitoring plots will provide more information of C stock changes, but few are available. More importantly, long-term monitoring plots obtained should be representative for the management intensities implemented in the study area. Using samples collected from highly managed plots for C uptake potential prediction could lead to enormously biased projection. Due to the lack of more detailed information about management implemented in each plantation (e.g. frequency, timing), we were unable to quantify each management practices' impacts on forest C dynamics. More efforts should be placed to design and obtain C stock information from long-term monitoring plots, though this is time-consuming and labor intensive.

5. Conclusions

Forest management greatly influenced forest biomass C increment in *Pinus*, *Cunninghamia*, and *Eucalyptus*. Although SOC differences were barely detectable among plantations under varied management groups,

vegetation C accumulation rates were generally larger under higher management intensities in *Pinus massoniana* and *Eucalyptus*, except for *Cunninghamia lanceolata*. Potential vegetation C accumulation enhancement is achievable by implementing higher management intensity, while cautious should be noticed to avoid sacrificing other ecological benefits. Moreover, we recommend that different management intensities should be considered for projection of future C sink from forest plantations, which is specifically important for regional or national estimations.

CRediT authorship contribution statement

Zhen Yu: Conceptualization, Methodology, Formal analysis, Writing - original draft. **Guoyi Zhou:** Methodology, Writing - review & editing. **Shirong Liu:** Conceptualization, Writing -review & editing. **Pengsen Sun:** Methodology, Validation, Writing - review & editing. **Evgenios Agathokleous:** Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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